Employee Attrition & Engagement

Advanced HR Analytics | 2026

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ABSTRACT

- Predicts employee attrition using data analytics
- Combines ML models and Power BI dashboards
- Enables proactive HR decision-making
- Employee attrition directly impacts cost, productivity, and morale
- This project uses HR analytics + machine learning
- Predicts employee attrition before resignation
- Provides actionable insights for HR leadership

OBJECTIVES

- Predict employee attrition
- Identify key attrition drivers
- Reduce hiring & training costs
- Support data-driven HR strategies
- Predict employee attrition accurately
- Identify key factors influencing attrition
- Enable proactive HR decision-making

INTRODUCTION

- Attrition impacts cost & productivity
- Traditional HR is reactive
- Predictive analytics enables early action
- Attrition causes knowledge and talent loss
- Traditional HR reports are reactive
- Predictive analytics enables early intervention
- Data-driven HR improves retention strategy

MOTIVATION

- High employee turnover
- Rising replacement cost
- Need for intelligent HR analytics
- High replacement and onboarding costs
- Need for predictive HR analytics
- Growing demand for HR data scientists
- Rising employee turnover in organizations

LITERATURE SURVEY

- Salary & satisfaction affect attrition
- ML improves prediction accuracy
- BI improves management decisions
- Job satisfaction and salary impact attrition
- Machine learning improves prediction accuracy
- Visualization improves management adoption
- Limited ML + BI integration in existing studies

EXISTING SYSTEM

- Manual Excel reports
- Reactive decision-making
- No predictive insights
- Manual Excel-based HR reports
- No attrition prediction mechanism
- Reactive decision-making
- Limited strategic insights

PROPOSED SYSTEM

- ML-based attrition prediction
- Automated HR analytics
- Interactive Power BI dashboards
- Machine learning-based attrition prediction
- Automated HR data analysis
- Interactive Power BI dashboards
- Proactive retention strategy

SYSTEM MODULES

- Data Collection & Preprocessing
- Exploratory Data Analysis (EDA)
- Machine Learning Model Development
- Power BI Visualization & Reporting

IMPLEMENTATION

- Python (Pandas, NumPy, Scikit-Learn)
- Jupyter Notebook (Google Colab)
- Machine learning models
- Power BI for dashboard creation

RESULTS & DISCUSSION

- High prediction accuracy
- Key attrition factors identified
- Improved HR decision support
- High attrition prediction accuracy
- Identified key attrition drivers
- Improved HR visibility through dashboards
- Supports data-driven HR planning

FUTURE SCOPE

- Real-time HRMS integration
- Explainable AI (SHAP)
- Employee sentiment analysis
- GenAI-powered HR chatbot

CONCLUSION

- Transformed HR data into predictive insights
- Reduced attrition risk proactively
- Supports strategic HR decision-making
- Industry-ready and scalable solution
- Predictive HR intelligence achieved
- Proactive attrition control
- Business-ready HR solution

FINAL OUTPUT

Our Values

Integrity

We uphold **honesty and transparency** in decisions.

Innovation

We encourage **creative thinking and adaptability** always.

Thank you!

Thank you for the constant support,
motivation, and trust throughout the journey.